

Roll No.....

MID SEMESTER EXAMINATION SEPTEMBER - 2025

Name of the Program: B. Tech Mechanical (R & A)

Semester: V

Name of the Course: ROBOTICS

Course Code: TME 513

Time: 90 minutes

Maximum Marks: 50

Note:

- (i) Answer all the questions by choosing any one of the sub questions.
- (ii) Each question carries 10 marks

Q1	(10 marks)	CO1
(a)	Define the following robot terms: i) payload, ii) workspace, iii) repeatability, iv) accuracy, and v) precision	
OR		
(b)	Discuss the Articulated and SCARA robot arm configurations with diagram?	
Q2	(10 marks)	CO1
(a)	Discuss the advantages and limitations of different robot configurations. Which configuration would be most suitable for high-precision assembly tasks and why?	
OR		
(b)	What are the essential components of a robot as a complete system?	
Q3	(10 marks)	CO2
(a)	Explain the types of mappings along with the diagrams.	
OR		
(b)	Explain the types of kinematic descriptions.	
Q4	(10 marks)	CO2
(a)	A robot's tool frame {T} is located relative to the base frame {B} such that the tool frame is rotated by 90° about the Z-axis of the base frame. Then it is translated by (2, 3, 5) units along X, Y, and Z axes of the base frame. Find the homogeneous transformation matrix.	
OR		
(b)	A point $P = (3, -2, 5)$ is rotated by 90° about the X-axis, and then translated by the vector $(-2, 4, 1)$. Using rotation and translation mapping, find the new coordinates of the point.	
Q5	(10 marks)	CO3
(a)	Define forward kinematics. Explain its significance in robotics and how it differs from inverse kinematics. Include examples of forward kinematics used in industrial robots.	
OR		
(b)	Consider a 2-DOF planar manipulator with: Link lengths: $L_1=2\text{m}$, $L_2=1.5\text{ m}$. Joint angles: $\theta_1=30^\circ$, $\theta_2=45^\circ$. Using forward kinematics, calculate the position of the end-effector in the base frame.	

